

BUSINESS PROCESS ENGINEERING

ASSIGNMENT: 03

SUBMITTED BY:

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ALAM

REG NO:

FA22-BSE-043

SEC: 0A

CASE STUDY :

PASSPORT OFFICE :

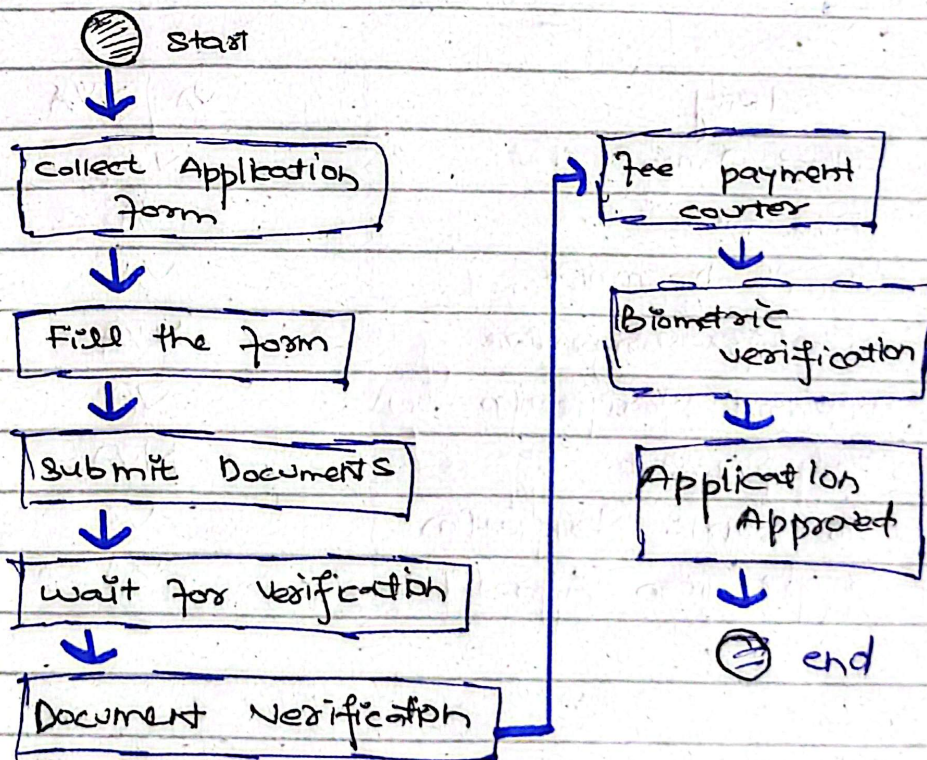
PROCESS: 01

PASSPORT APPLICATION :

Objective :

To Issue a passport to an applicant by verifying personal documents, collecting biometric data, and approving the application efficiently.

AS-IS PROCESS FLOW



FLOW - TIME ANALYSIS

Activity	Time (min)
1 collect application form	10
2- Fill form	20
3- Submit documents	10
4- wait for verification	30
5- Document verification	15
6- Fee Payment	20
7- Biometric Verification	10
8- Application Approved	5
Total cycle Time	120 min

Formula:

$$\begin{aligned}
 \text{Total cycle Time} &= \sum \text{Activity Times} \\
 &= 10 + 20 + 10 + 30 + 15 + 20 + 10 + 5 \\
 &= 120 \text{ min}
 \end{aligned}$$

VALUE ADDED ANALYSIS (VA)

Activity	VA / NVA
collect form	NVA
Fill form	VA
Submit documents	VA
wait for verification	NVA
Document verification	VA
Fee payment	NVA
Bio metric Verification	VA
Application Approval	VA

Value Added Time

$$20 + 10 + 15 + 10 + 5 = 60 \text{ min}$$

Non-Value-Added Time

$$10 + 30 + 20 = 60 \text{ min}$$

PROCESS EFFICIENCY:

$$\text{Process Efficiency} = \frac{\text{VA Time}}{\text{Total Time Cycle}} \times 100$$

$$= \frac{60}{120} \times 100$$

$$= 50\%$$

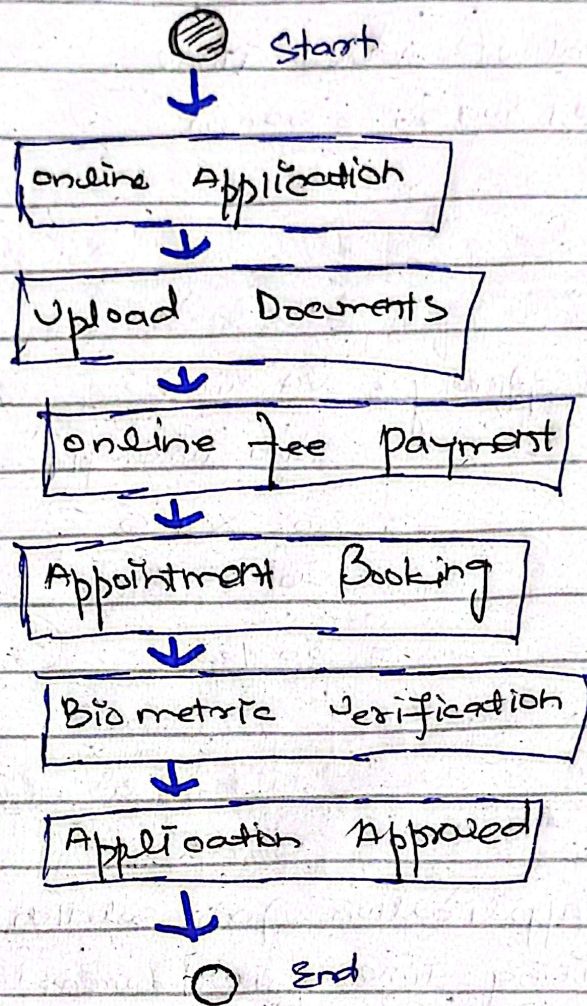
PROBLEMS IDENTIFIED

- manual Application form selection
- Long waiting Time for document verification.
- Separate counter for fee payment
- multiple manual steps increase processing time.

IMPROVEMENT PROPOSED:

- online application submission
- Digital Document upload
- online fee payment
- Appointment - Based Biometric verification
- Automated document verification before office visit.

To-Be Process Flow:



Improved Flow Time Analysis	
Activity	Time (min)
① Online Application	10
② Upload Documents	5
③ online fee payment	5
④ Appointment waiting	10
⑤ Biometric Verification	10
⑥ Application Approved	5
Total Cycle Time	45 min

NEW PROCESS EFFICIENCY

Value Added Time

$$10 + 5 + 5 + 10 + 5 = 35 \text{ min}$$

NVA Time

10 min

$$\begin{aligned} \text{Process Efficiency} &= \frac{35}{45} \times 100 \\ &= 77.8\% \end{aligned}$$

EFFICIENCY IMPROVEMENT

$$\text{Efficiency Improvement} = \frac{\text{new Efficiency} - \text{old E}}{\text{old Efficiency}} \times 100$$

$$= \frac{77.8 - 50}{50} \times 100$$

$$= 55.6\%$$

Final Result

$$\text{old Efficiency} = 50\%$$

$$\text{new efficiency} = 77.8\%$$

$$\text{Efficiency Improvement} = 55.6\%$$

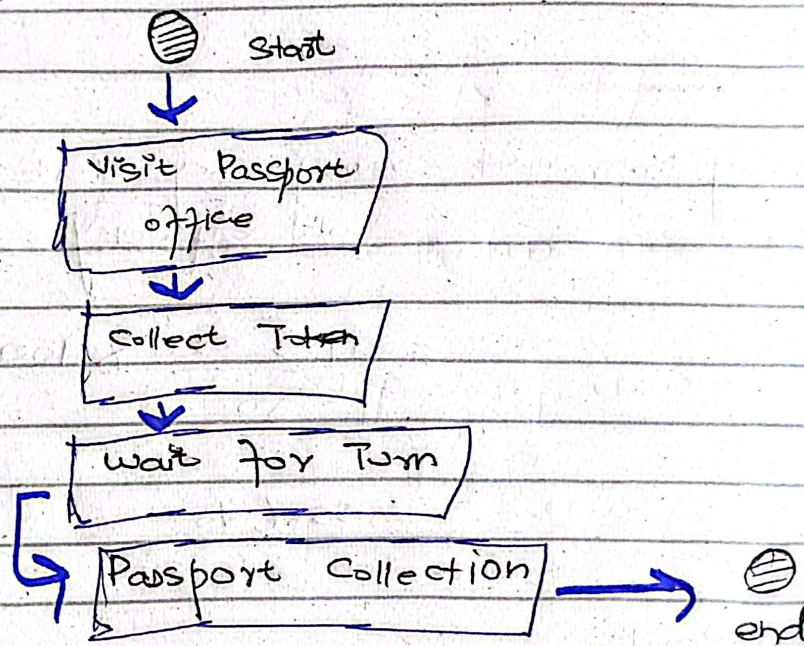
PROCESS: 02

PASSPORT COLLECTION:

Objective :

To deliver the issued passport to the applicant after verifying their identity in a quick and secure manner.

AS-IS PROCESS FLOW:



FLOW TIME ANALYSIS:

Activity	Time (min)
Visit Passport office	20
Collect Token	10
wait for Turn	40
Identity Verification	10
Passport Collection	5

Total cycle Time = 85 min

VALUE ADDED ANALYSIS:

ACTIVITY	VA / NVA
visit passport office	NVA
collect Token	NVA
wait for Turn	NVA
Identity verification	VA
Passport collection	VA

Value Added Time = 10 + 5 = 15 min

NON - VA = 20 + 10 + 40
= 70 min

PROCESS EFFICIENCY

$$\text{Process Efficiency} = \frac{15}{85} \times 100$$

$$= 17.6\%$$

PROBLEMS IDENTIFIED:

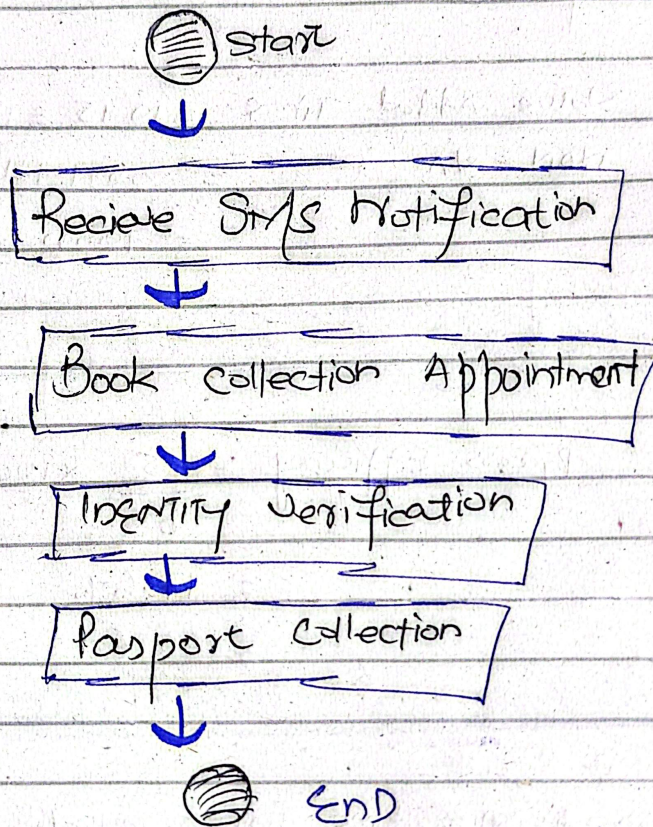
- o Long waiting time before collection
- o manual Token Distribution
- o No scheduled Appointment System.
- o Overcrowding at the passport office.

next page →

IMPROVEMENT PROPOSED:

- SMS Notification when passport is ready
- Online Appointment for collection.
- Digital queue management system.
- Faster identity verification using QR code or biometric scan.

TO-BE PROCESS FLOW:



IMPROVED FT- ANALYSIS:

	min()
Receive SMS Notification	2
Book Appointment	5
Identity Verification	8
Passport Collection	5
Waiting at office	2

Total Cycle Time = 21 min

NEW PROCESS EFFICIENCY:

Value - Added Time

$$1 + 5 + 8 + 5 = 19 \text{ min}$$

NVA Time

2 min

Formula

$$\begin{aligned} \text{Process Efficiency} &= \frac{19}{21} \times 100 \\ &= 90.5\% \end{aligned}$$

Efficiency Improvement:

$$\begin{aligned} \text{Efficiency Improvement} &= \frac{(\text{New E} - \text{old E})}{\text{old Efficiency}} \times 100 \\ &= \frac{90.5 - 17.6}{17.6} \times 100 \\ &= 414.2\% \end{aligned}$$

Final Results

• Old Efficiency	= 17.6%
• New Efficiency	= 90.5%
• Efficiency Improvement	= 414.2%